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A Review on Secure Data Deduplication: Cloud Storage Security Issue

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ABSTRACT

Cloud storage service providers caters to the need of organizations and individuals by allowing them to store, transfer and backup their ever-increasing amount of data at low cost along with providing access to the other resources of cloud. For providing efficient data storage, cloud service providers utilize most widely employed deduplication technique as it allows storage of single instance of data and removes duplicate copies of data, thus mitigating storage overhead and saving upload bandwidth. Clients uploading their data on cloud are most concerned about the security, integrity, privacy and confidentiality of their data. Conventional encryption usually employed to encrypt data while outsourcing it, is not recommended as it conflicts with data deduplication technique and so in most cases, Convergent Encryption (CE) and Proof of Ownership (PoW) are used to protect confidentiality and integrity of data. Several other approaches such as Provable Data Possession (PDP), Proof of Retrievability (POR), secure keyword search, DupLESS, Proof of Storage with Deduplication (PoSD), Dekey, Message-Locked Encryption, Attribute Based Encryption (ABE) and Identity Based Encryption (IBE) have been researched to address client's security concerns and this paper does a literature review on such various proposed approaches for secure deduplication techniques in cloud storage.

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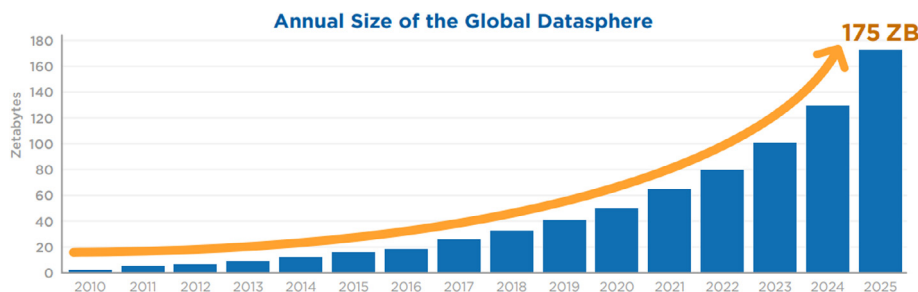


Fig. 1. Annual Size of the Global Datasphere.

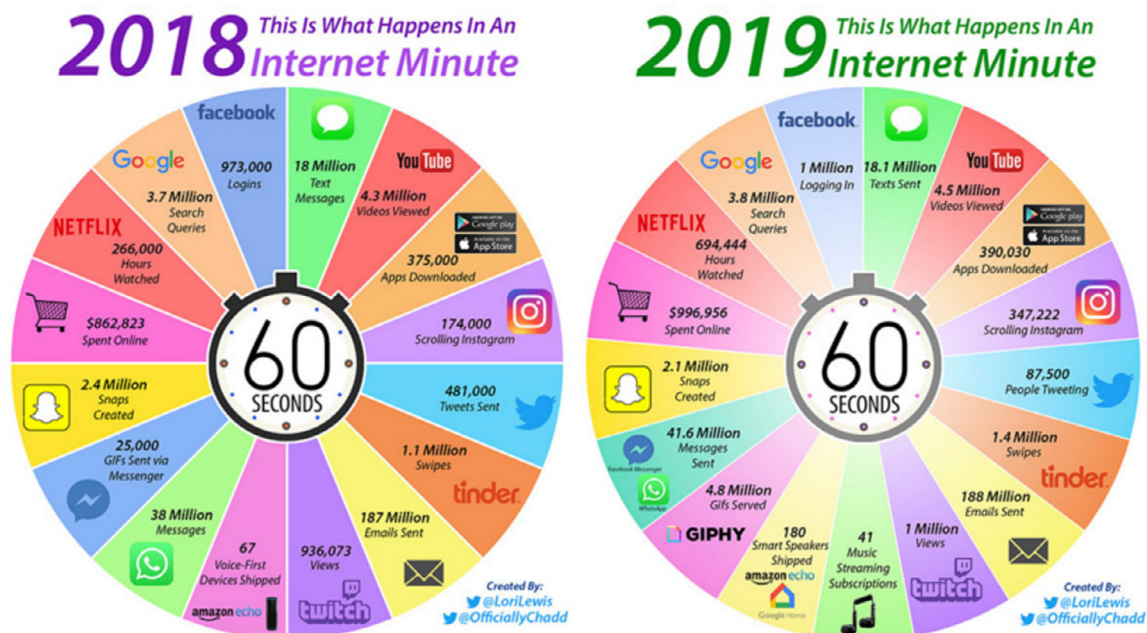


Fig. 2. Internet Minute Infographic (2018–2019).

1. Introduction: Massive Growth of Data

Internet users have increased dramatically in the past few years. More and more people are coming online. It has become an important part of their daily life and connected with their social lives. Besides, because of this COVID pandemic, the way we interact with the internet has fundamentally changed. We are using technology to do things we have never done before. Greater number of jobs, businesses, educational institutions are taking benefit of technology to get the work done even in such hard times. The changes it has brought are here to stay, but such advancements in technology also expose it to some serious security threats which makes cyber threats one of the biggest global risks. In the past few weeks of pandemic online threats have risen as much as six times. Attackers try to steal or modify our data and can even take control of our systems. We have examples of global-scale attacks like ransomware and some attacks on the zoom platform.

There has been tremendous growth in usage of computers, information, online applications, mobile computing in recent years. This has resulted into exponential growth of user base and their data across the globe. Ever-increasing data and storage space required for storing that data has become a prime concern. People send out their data to remote storage due to limited storage capacity, overhead and maintenance and therefore less fiscal hence cloud computing becomes more popular.

As per International Data Corporation (IDC Report, 2020) report, Global Datasphere is the combination of data generated, captured or replicated through the digital content from all over the world. As shown in Fig. 1, IDC predicts that the Global Datasphere will grow from 33 Zettabytes (ZB) (1 ZB = 10^{21} Bytes or 2^{70} Bytes) in 2018 to 175 ZB by 2025.

As shown in Fig. 2 by -2019 Internet Minute Infographic (2019) and as shown in Fig. 3 by Internet Minute Infographic (2020), we can visualize the use of cloud storage of various cloud applications.

2. Security issues in Cloud Computing

2.1. Cloud computing

Efficient utilization of remote resources is called cloud computing in very simple words. As per the NIST Definition Mell and Grance (2011) "Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

2020 This Is What Happens In An Internet Minute



Fig. 3. Internet Minute Infographic (2020).

Table 1
Number of breaches per month in the year 2019

Month	Total Breaches	Data Records Breaches
January	59	1.1 Billion
February	58	870 + Million
March	79	2.1 Billion
April	71	1.3 Billion
May	77	1.3 Billion
June	64	39 Million
July	92	2.3 Billion
August	94	105 Million
September	75	531 Million
October	110	421 Million
November	87	1.3 Billion
December	90	627 Million

2.2. Cloud computing security issues

As per [Cyber Security Breaches Report of Black Hat Ethical Hacking \(2019\)](#), 12.3 Billion was total number of records were breached and as report 3,83,365 US Dollar was the cost per single attack in 2019. [Table 1](#) gives a detail of total 956 major data breaches reported in 2019. As per [Businesses Report](#) in year 2019: 48% of businesses identified at least 1 attack per month and 62% of businesses can respond to a breach immediately. At end of 2019, every 30 seconds business fell victim to a ransomware attack and it is rises to every 11 seconds by 2021. And also organizations report in year 2019 says: 67% of organizations are reported being breached at some of point in the past systems and also 31% of organizations experienced of cyber-attack on their own operation infrastructure. Unfortunately, 50% of them have not updated their security strategy from 3 year and more. If we talk about types of attack than as per report in year 2019: 80% attacks from phishing attacks and 28% come from spoofed mail attacks and likely 27% attack came from malware, ransomware and spyware related attacks. Average in 197 days breach discover by companies and

which companies discover within 30 days, saved more than 1 million USD compared to those companies which took 197 days.

As per [Verizon Data Breach Investigations Report \(2020\)](#), total 3,950 data breaches reported across worldwide and continue. List of these breaches according to sector and service are as shown in the report. As per this report of year 2020 till date there is 70% external actors behind breaches, organized criminal groups were behind 55% of breaches, 30% involved internal actors, only 4% of breaches had four or more attacker actions, 1% involved partner actors and 1% featured multiple parties. If we talk about victims there were 81% of breaches contained in days or less, 72% of breaches involved large business victims, 58% of victim had personal data compromised, 28% of breaches involved small business victim. There are other Commonalities main 86% of breaches involved in financially motivated, 43% of breaches involved in web application, 37% of breaches stole or used credentials, 27% of malware incidents were ransomware, 22% of breaches involved phishing.

As per [Sehgal et al. \(2020\)](#), *Data Movement*: it means generally the data are being sent from client to server and vice versa so the security of the data is the concern while data is moving from source to destination. *Loss of Control*: once the data is uploaded in the cloud, the privacy of the data is the major concern because cloud service provider or the employees can misuse the data for advertisement or personal benefits. *Uncertain Performance*: As the resources are shared among multiple users there might be the issues related to the uncertain performance. *Identity Authentication and Unauthorized Access*: it is very difficult to identify and verify between genuine users and fake users/attackers due to different cyber security attacks. *Data Theft*: it can be possible if cloud service providers are not updated with proper storage compliance. *Denial of Service Attacks*: it can be possible if cloud service providers are not updated with detect and defence system of cyber attacks. *Data Integrity*: it is very critical if cloud service providers are changing the data of the users. *Invasion of Privacy and Activity Monitoring*: it might be possible by providing some free services cloud service providers are collecting the user confidential data and sharing or monitoring the activities of the users.

Apart from that, “Web Threat Models: HTML, Rendering Content, Remote Scripting, Cookies, Frames and frame busting; Web Application Security: Injection, Broken Authentication and Session Management, Cross-Site Scripting (XSS), Insecure Direct Object References, Security Misconfiguration, Sensitive Data Exposure, Missing Function Level Access Control, Cross-Site Request Forgery (CSRF), Using Components with Known Vulnerabilities, Un-validated Redirects and Forwards” are additional security considerations for cloud computing as per discussion in [Sehgal et al. \(2020\)](#).

Data encryption technologies like Identity Based Encryption, Attribute Based Encryption, Homomorphic Encryption, and Searchable Encryption are discussed in [Yang et al. \(2020\)](#) related to cloud storage.

As per [Hashizume et al. \(2013\)](#), “A threat is a potential attack that may lead to a misuse of information or resources, and the term vulnerability refers to the flaws in a system that allows an attack to be successful. Insecure interfaces and APIs, Unlimited allocation of resources, Data-related vulnerabilities, Vulnerabilities in Virtual Machines, Vulnerabilities in Virtual Machine Images, Vulnerabilities in Hypervisors, and Vulnerabilities in Virtual Networks are vulnerabilities in cloud computing. Account or service hijacking, Data scavenging (since data cannot be completely removed from, unless the device is destroyed), Data leakage, Denial of Service, Customer-data manipulation, VM escape, VM hopping (VM is able to gain access to another VM), Malicious VM creation, Insecure VM migration, and Sniffing/Spoofing virtual networks” are threats in cloud computing.

Table 2
Critical Security Control Services Provided by Major Cloud Services Providers

Critical Security Control Services	AWS (2020)	Azure (2020)	Cloud (2020)	Oracle (2020)	IBM (2020)	Alibaba (2020)
Firewall & ACLs	✓	✓	✓	✓	✓	✓
IPS/IDS	3rd Party	✓	3rd Party	3rd Party	3rd Party	✓
Web Application Firewall (WAF)	✓	✓	✓	✓	✓	✓
SIEM & Log Analytics	✓	✓	✓	✓	✓	✓
Antimalware	3rd Party	✓	3rd Party	3rd Party	3rd Party	✓
Data Loss Prevention (DLP)	✓	✓	✓	3rd Party	3rd Party	✓
File Integrity Monitoring (FIM)	3rd Party	✓	3rd Party	3rd Party	3rd Party	3rd Party
Key Management	✓	✓	✓	✓	✓	✓
Encryption At Rest	✓	✓	✓	✓	✓	✓
DDoS Protection	✓	✓	✓	✓	✓	✓
Email Protection	3rd Party	✓	✓	3rd Party	3rd Party	3rd Party
SSL Decryption & Reverse Proxy	✓	✓	✓	3rd Party	✓	✓
Endpoint Protection	3rd Party	✓	3rd Party	3rd Party	3rd Party	✓
Certificate Management	✓	✓	3rd Party	3rd Party	✓	✓
Container Security	✓	✓	✓	✓	✓	✓
Identity and Access Management	✓	✓	✓	✓	✓	✓
Privileged Access Management (PAM)	3rd Party	✓	3rd Party	3rd Party	3rd Party	3rd Party
Multi-Factor Authentication	✓	✓	✓	✓	✓	✓
Centralized Logging & Auditing	✓	✓	✓	✓	✓	✓
Load Balancer	✓	✓	✓	✓	✓	✓
LAN	✓	✓	✓	✓	✓	✓
WAN	✓	✓	✓	✓	✓	✓
VPN	✓	✓	✓	✓	✓	✓
Governance Risk and Compliance Monitoring	✓	✓	✓	3rd Party	3rd Party	✓
Backup and Recovery	✓	✓	✓	✓	✓	✓
Vulnerability Assessment	✓	✓	✓	✓	✓	✓
Patch Management	✓	✓	3rd Party	✓	3rd Party	3rd Party
Change Management	✓	✓	3rd Party	3rd Party	3rd Party	✓

As per [Rao et al. \(2016\)](#), Data privacy and Integrity, Data recoverability and vulnerability, Improper media refinement, and Data backup are cloud storage issues in cloud computing. As per [Zhang et al. \(2020a\)](#), “the following topics remain open research issues that should be further investigated. Secure Data Integrity Verification from Smart Contract, Combination of Encrypted Deduplication and Symmetric key Searchable Encryption.”

2.3. Cloud Data Storage Providers

Now a days remote data storage is one of the majorly used applications of cloud computing. Security is certainly critical for any size of organization and users of cloud storage. Cloud computing storage application must provide highly available access to data while maintaining high performance and maximum scalability. In addition, reliability in a storage application is of the utmost importance and the correctness of data must be guaranteed. Cryptography of some type is typically the security solution, but it is the where and when of that concept that becomes important. At a minimum, client wants its data in the cloud to be secure and intact. The Cloud Service Provider (CSP) hosting client's data needs to provide access to data, and the data can't be read or modified by unauthorized user.

Cloud data storage offer a huge pool of shared resources, so users move it to meet their needs. Cloud storage integrates different kinds of storage devices together through application cluster, web technology, and distributed file system to provide users with storage service. To name some of the service providers –Amazon Drive, Apple iCloud, BigMIND, CertainSafe Digital Safety Deposit Box, Cloud Me, Crash Plan, DriveBox, Dropbox, Google Drive, IBackup, IDrive, Microsoft OneDrive, Mega, Mozy, NextCloud, pCloud, SpiderOak, SugarSync, Sync.com, Team Drive, Ubuntu One, Wuala are some of Cloud storage providers mentioned above offer cloud storage with various types of security Confidentiality, Integrity and Availability.

As per [Grigorof and Mocanu \(2019\)](#) critical cloud security controls provided by major cloud services providers are shown in [Table 2](#) for greater understanding purpose. It is observed based on [Table 2](#); if situation is to manage full critical security controls Azure can be given high priority. If the possible situation is related to the concern related to email protection of any reputed organization then Google Cloud can be given high priority after Azure. If situation is like to manage the services like backup and recovery, vulnerability assessment, patch management and change management in your application then Azure or AWS can be given high priority. If situation is to manage only firewall, log analytics, key management, encryption at rest, DDoS protection, identity & access management and multi-factor authentication only then any of the given cloud providers can be used.

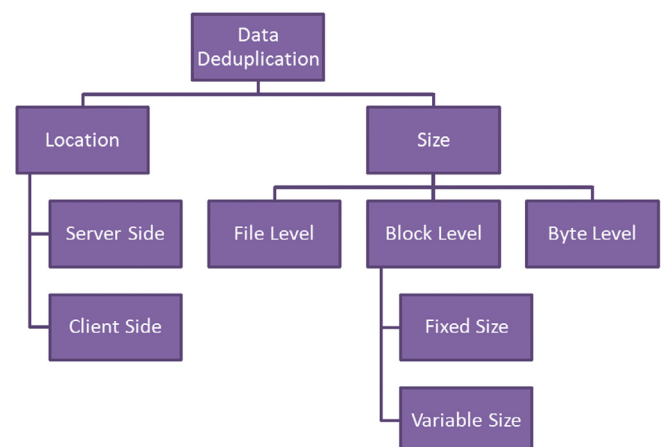


Fig. 4. Data Deduplication.

Table 3

Comparison of various Data Deduplication schemes by Confidentiality, Integrity and Deduplication type and Deduplication level

Authors	Confidentiality	Integrity	Deduplication Type	Deduplication Level
Bai et al. (2020)	AES-128	Yes	Server Side	Block Level
He et al. (2020)	Elliptic Curve	Yes	Client Side & Server Side	File Level
Premkamal et al. (2020)	Yes	Yes	Server Side	File Level
Lee and Park (2020)	Yes	SHA-1	Server Side	File Level & Block Level
Li et al. (2020)	AES-256	SHA-256	Client Side	Block Level
Li et al. (2020)	AES	SHA-1	Client Side	File Level & Block Level
Liu et al. (2020)	AES	Yes	Server Side	File Level
Nayak and Tripathy (2020)	AES	Yes	Server Side	File Level
Shen et al. (2020)	Yes	MAC	Server Side	File Level & Block Level
Yin et al. (2020)	No	SHA-1	Client Side & Server Side	File Level & Block Level
Yuan et al. (2020)	AES-128 & AES-256	SHA-128 & SHA-256	Server Side	Block Level
Wang et al. (2020)	Yes	Yes	Client Side	File Level
Zhang et al. (2020b)	No	SHA-1	Server Side	Block Level
Ni et al. (2018)	AES	SHA-256	Client Side	File Level
Prajapati et al. (2017)	No	SHA	Client Side	File Level
Shin et al. (2017)	AES-256	SHA-256	Server Side	File Level
Scanlon (2016)	No	Yes	Server Side	File Level
Li et al. (2015)	Yes	Yes	Client Side & Server Side	File Level & Block Level
Duan (2014)	AES	Yes	Server Side	File Level & Block Level
Prajapati and Shah (2014)	Blowfish	SHA-256	Client Side	File Level
Stanek et al. (2014)	AES-256	SHA-256	Client Side	File Level
Keelveedhi et al. (2013)	AES	SHA-256	Server Side	File Level
Li et al. (2013)	AES-256 with CBC	SHA-256	Client Side	File Level & Block Level
Puzio et al. (2013)	Yes	SHA-256	Server Side	Block Level
Meyer and Bolosky (2012)	No	MD5	Server Side	File Level & Block Level
Zheng and Xu (2012)	Yes	Yes	Client Side	File Level
Rahumed et al. (2011)	AES-128	SHA-1	Client Side	File Level
Storer et al. (2008)	Yes	HMAC	Client Side	Block Level

2.4. Cloud Data Storage Security Issue: Data Deduplication

Despite of various choices available for data storage including cloud data storage, one of the major challenges faced by users & organizations is about data duplication. It has been observed that for a single user or single transaction, there is a lot of duplication of data resulting due to usage of different sources of information. Data deduplication is one of the effective techniques for data reduction. This technique ensures storage one single copy of each data. This is possible by comparison of data fingerprints with the existing stored data and thus identifies duplicate data.

Table 4

Various Notations.

Notation	Meaning
H	Hash of Data
Mul	Group Multiplication
Exp	Group Exponentiation
U	User
KS	Key Server
CSP	Cloud Service Provider
SE	Symmetric Encryption
ASE	Asymmetric Encryption
CuE	Customized Encryption
Pair	Pairing
F	File
B	Block
$ G_i $ and $ G_T $	Size of group element in G_i and G_T
N	Number of Key Server
t	Number of key share server (kss)
LDP	Leakage by Deduplication Pattern
DR	Duplicate Reply
FA	Frequency Attack
OffBF	Offline Brute Force Attack
OnBF	Online Brute Force Attack
CompKS	Compromise of KS
LeHT	Leakage from Hash or Tag
DP	Data Privacy
DI	Data Integrity
CR	Collusion Resistance

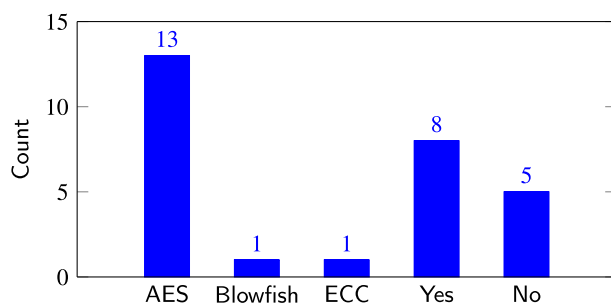
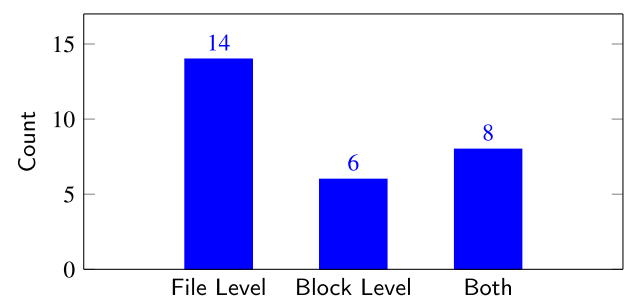
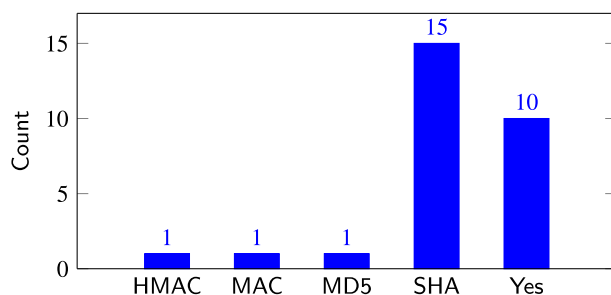
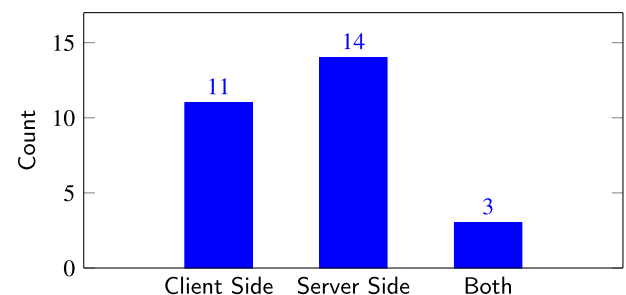
Due to technological advancement, the volume of data owned by any organization and individual is constantly increasing and this has led to the growing demand for storing this ever-increasing amount of data. This demand of storing huge volume of data cannot be satisfied by conventional data storing methods which used physical storage such as flash drives and hard disks. These storage devices provide limited storage capacity and have other limitations also. As a result, the modern technological world has turned towards cloud storage to meet their needs of outsourcing large volume of data and files because cloud storage provides certain facilities like: giving storage space according to user's needs at low budget, clients do not have to incur maintenance overhead, it allows users to backup and recover their data, and provides a shared pool of resources for users to process and transfer their files seamlessly. For storing user's data efficiently, cloud storage providers deploy 'Deduplication Technique'.

Deduplication reduces storage space as it allows only single instance of data to be retained and removes all the duplicate copies of data files and all the redundant data files are replaced by pointer which points to the unique instance of data. Data Deduplication technique optimizes the storage space of cloud storage service providers thus leading to efficient use of disk space. Another benefit of deduplication is that, if it is done at source then the duplicate data copies will not be transmitted and save upload bandwidth. Data deduplication tends to manage increasing volume of data and it also reduces operational cost and cost of storage. As per Microsoft study by Meyer and Bolosky (2012), 87% of storage can be achieved for backup images by efficiently using block/chunk size in block-level Data Deduplication. As per the work done by Zhang et al. (2020b), Data Deduplication can be achieved up-to 2.9 to 16.9X with efficient use with Data Compression. As per the work done by Ni et al. (2018), secure Data Deduplication can be also used effectively in fog computing to reduce the communication overhead of mobile users and fog nodes. Primarily deduplication can be done based on size: at file level, block level, byte level client and based on location: Client Side or Server Side Deduplication as shown in Fig. 4.

Table 5

Comparison of various Data Deduplication schemes by Implementation, Key Generation Cost, Encryption/Hash Cost, Ciphertext Size, Tag Size and Security Analysis

Authors	Implementation	Key Generation Cost	Encryption/Hash Cost	Ciphertext Size	Tag Size	Security Analysis
Bai et al. (2020)	C	KS: H + Mul + Exp	H + SE + Pair + Mul + Exp	$ B + G_T + G $	$ G $	OnBF, CompKS, LeHT, DP & DI
He et al. (2020)	C++	KS: H + Exp	U: ASE + H + Mul + Pair + Exp KS: Exp	$ F + G_T + G $	$ G $	OnBF, LeHT, DP & DI
Premkamal et al. (2020)	Python	U: H + Mul + Exp	U: ASE + Exp + Mul + Pair	$ F $	$ SHA $	DP & DI
Lee and Park (2020)	C++ & Solidity	H	CuE	$ F + B $	$ SHA $	DP & DI
Li et al. (2020)	C++	H	SE	$ B $	$ SHA $	FA & OfBF
Li et al. (2020)	C++	H	SE + CuE	$ F + B $	$ SHA $	FA, DP & DI
Liu et al. (2020)	C	H + Mul + Exp	$(2t+3)SE \cdot G + Exp + Mul + Pair$	$3(t+3) G_T + F $	$ G $	LeHT, DP, DI & CR
Nayak and Tripathy (2020)	C	H + 2Mul + 6Exp	SE + 6Exp + 2Mul + Pair	$ F + 2 G + G_T $	$ G $	OfBF, OnB, FLeHT & DI
Shen et al. (2020)	C	H + 2Mul + 2Exp	SE + 2Exp + 2Mul	$ F + B $	$ MAC $	OfBF & DI
Yin et al. (2020)	Detail Not Available	–	H	–	$ SHA $	DI
Yuan et al. (2020)	Java & Solidity	H + 2Mul + Exp	SE + 2Exp + Mul + Pair	$ G_1 + G_2 + G_T $	$ SHA $	DP & DI
Wang et al. (2020)	Java	H + 6Mul + 6Exp	ASE + 6Exp + 6Mul + Pair	$2 G + 2 G_T $	$ G_1 + G_2 $	CompKS, DP & DI
Zhang et al. (2020b)	C	–	H	$ B $	$ SHA $	–
Ni et al. (2018)	C & C++	H + Mul + Exp	SE + Exp + Mul + Pair	$ F + G + G_T $	$ SHA $	LDP, DR & OnBF
Prajapati et al. (2017)	Java	–	H	$ F $	$ SHA $	DI
Shin et al. (2017)	Python	U: H + 2(t+1)Mul + 4Exp kss: 2Exp	U: SE + (N+1)Exp + Mul + Pair	$ F + G_T + N \cdot G $	$ G $	OnBF, OfBF, CompKS, LeHT & CR
Scanlon (2016)	XML	–	H	$ F $	$ G $	–
Li et al. (2015)	Python	H	SE	$ F + B $	$ G $	DP & DI
Duan (2014)	Java	U: H + 2(t+1)Mul + 2Exp KS: Exp	U: SE + H	$ F $	$ HMAC $	LDP & CompKS
Prajapati and Shah (2014)	Python	U: H + 2Mul + 2Exp KS: Exp	U: SE + H	$ F $	$ HMAC (SHA) $	OnBF, CompKS, DP & DI
Stanek et al. (2014)	C++	H + Mul + Exp	SE + H + Exp + Mul + Pair	$ F + G_1 + G_2 + G_T $	$ SHA $	LDP, DP & DI
Keelveedhi et al. (2013)	Python	U: H + 2Mul + 2Exp KS: Exp	U: SE + H	$ F $	$ HMAC (SHA) $	OnBF, CompKS, DP & DI
Li et al. (2013)	C & C++	H	SE + H	$ F + B $	$ SHA $	DP & DI
Puzio et al. (2013)	Hardware Security Module	H	SE + H	$ F + B $	$ SHA $	OnBF, CompKS, DP & DI
Meyer and Bolosky (2012)	C & C++	–	H	$ F + B $	$ MD5 $	–
Zheng and Xu (2012)	Detail Not Available	H + Mul + Exp	SE + H + Exp + Mul	$ F $	$ G $	CompKS, DP & DI
Rahmed et al. (2011)	C	H	SE + H	$ F $	$ SHA $	DP & DI
Storer et al. (2008)	Detail Not Available	H	SE + H	$ B $	$ HMAC $	CompKS, DP & DI

**Fig. 5.** Confidentiality algorithms used in various work.**Fig. 7.** Deduplication type used in various work.**Fig. 6.** Integrity algorithms used in various work.**Fig. 8.** Deduplication level used in various work.

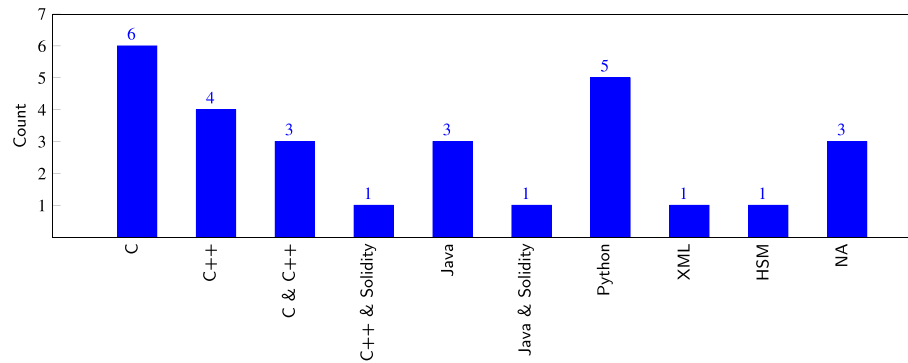


Fig. 9. Implementation detail of various Deduplication work done by Authors.

3. Literature Review of Existing Data Deduplication

According to a recent IDC study [DuBois et al. \(2011\)](#), almost 80% of corporations surveyed indicated that they were exploring data deduplication technologies in their storage systems to reduce redundant data and thus increase storage efficiency and reduce storage costs. Data deduplication is a technique to reduce the amount of storage space for each entity to save its data. There are several copies of the same data being stored in the same place or the same piece of data in multiple places. Deduplication eliminates these extra copies which are the same, just to save a copy of the data.

Data privacy, security, confidentiality and integrity are few main concerns of storing data on cloud and to address these concerns, Convergent Encryption is used along with deduplication technique. Deduplication does not work with traditional encryption because various users will encrypt the data with their own unique keys resulting in different cipher texts for same data. Thus, Convergent Encryption is used as it allows deduplication and also provides data confidentiality. It computes cryptographic hash value of the data copy and uses the same hash value as a convergent key for the encryption of the data. Similar data copies produce identical convergent keys and hence similar cipher text, thus allowing the cloud storage to execute deduplication. Comparison of various Deduplication schemes by Confidentiality, Integrity and Deduplication type and Deduplication level are discussed in the [Table 3](#) for reference. As per the literature review in various approaches for providing secure deduplication in cloud storage are discussed. The different schemes proposed can be employed with existing cloud storage services. Convergent Encryption is majorly used as per the review.

Various notations are shown in the [Table 4](#) used by the [Table 5](#). The comparison of various deduplication schemes by key generation cost, encryption/hash cost, ciphertext size and various security analysis like against various attacks like online and offline brute force attack, compromise of key server, leakage from hash and collision resistance etc. are discussed in the [Table 5](#) for reference.

Different confidentiality and integrity algorithms used in various work are shown in [Fig. 5](#) and [Fig. 6](#). Different Deduplication type and Deduplication level used in various work are shown in [Fig. 7](#) and [Fig. 8](#). Based on the observations mostly AES is used for confidentiality, SHA is used for integrity and server side with file level Deduplication is used. [Fig. 9](#) provides the implementation detail of various Deduplication work done by authors.

3.1. Convergent Encryption based Data Deduplication

Although Convergent Encryption (CE) is used along with deduplication, it is possibly vulnerable to certain attacks such as:

Dictionary attack, Confirmation of File (COF) and Learn-the-Remaining-Information (LRI). To solve these issues of CE and to ensure that the benefits of both deduplication and CE are retained, [Puzio et al. \(2013\)](#) proposes ClouDedup. This solution provides data confidentiality and user's privacy without affecting the efficiency of deduplication. ClouDedup aims at block-level deduplication and contains two main components: a server – which will provide an additional security layer and a Metadata Manager (MM) – which helps in key management of blocks along with deduplication operations. By adding server and MM to the basic storage provider, the weaknesses of CE are overcome and data confidentiality is also ensured. Also by using these two additional components, the system is secured from single point of failure and the whole system will not fail even if a single component is compromised.

ClouDedup can be easily integrated with any basic cloud storage service provider. It also provides security against offline dictionary attacks as the server is not entrusted for the confidentiality of the data which is stored in cloud storage. On evaluating the proposed system, it is found that the overhead incurred by the MM is negligible and has no effect on space saving by deduplication. Also, the analysis of computational complexity depicts that the overall cost for storage and retrieval operations is linear thus making the metadata management scalable and also making system extensible to very large datasets. The limitation of the proposed system is that it might be vulnerable to online dictionary attack if the attacker manages to compromise one or more users. But as the server is not compromised, the data of other users except the compromised ones remains safe. Moreover, this attack can be mitigated by setting threshold limit for the rate with which users can send requests. ClouDedup can be further extended to include more security features and the proposed system could be optimised in terms of computation, bandwidth and storage space.

Two different models are proposed as solutions: authenticated and anonymous in [Storer et al. \(2008\)](#). Both these approaches utilize the convergent encryption which allows deduplication along with encryption. The proposed system consists of three main components: the client, the metadata store – which contains all the information necessary for users to reconstruct files from their respective chunks and the chunk store – which stores all the data blocks and gives these data blocks to users based on the ID on user's request. For both models, the chunking of data and its encryption takes place on client side. In the authenticated model, general convergent encryption is used along with which certain assumptions are made related to encryption keys and its management methods that are available to users. Whereas, the anonymous model's main aim is to conceal the user's and reader's identity. This model works on assumption that authentication is not needed as the encrypted data is protected from the attackers who do not have the right encryption key. For both these models, a map is made for individual file which gives information on how to rebuild a file

from its data blocks. From the security point of view, both models are evaluated and it was found that the system has negligible threat from external adversaries and the security risks from the inside adversaries are reduced on a large scale as chunking and encryption of data takes place on client side. This paper also studied the effects on system if an encryption key is compromised and it was found that loss in client's encryption key results in a severe security issues. Thus, the proposed system shows that deduplication and security can be combined to provide various security properties and for future work, the system can be extended to increase storage efficiency by adopting various parameters for chunking according to the type of files.

3.2. Data Deduplication based on file Popularity

Stanek et al. (2014) addresses the issue of conflict between the traditional encryption and deduplication. Unlike all the related work which presumes that all files need equal security, this paper proposes a scheme which provides security to the data on basis of their popularity. The files are divided in two categories: Popular and Unpopular. Initially, all the files are considered as unpopular files and they are encrypted using two different layers of security; the innermost layer is secured using convergent encryption while the outer layer is secured using the semantically secured cryptosystem. Later when the popularity threshold for unpopular files is reached, the outer security layer is removed and the file becomes popular and only the inner and weaker layer of convergent encryption remains. For extending the security and for indexing, two trusted entities are added to the traditional storage provider; they are: an identity provider and indexing service.

In this proposed system, encryption is done on client side whereas the decryption occur independent of the client. Security and performance evaluation of proposed system is also done. The protocols of the new system are proved to be secure under Diffie-Hellman assumption and the performance evaluation showed that overhead incurred by system setup, file upload, file access and file transition is negligible as compared to the resources consumed by the convergent and symmetric encryptions. In this novel encryption scheme, complete privacy is provided only for those users who upload unpopular files while for the users uploading popular files the privacy is weaker and it has scope of improvement in future. Better flexibility of popularity threshold and deletion of data in the given scheme are also included as future perspective. In this novel encryption system, semantic security is provided for all the unpopular data whereas for popular data the security is slightly weaker but a good bandwidth and storage is provided for popular data.

3.3. Dekey: Efficient use of file level and block level Deduplication

Li et al. (2013), first introduces baseline approach in which convergent key is encrypted with the help of master key which in turn is stored locally by each user and the encrypted convergent key is stored on cloud storage with its corresponding data copy. The main problem with the baseline approach is: key management creates storage overhead with increase in number of users and user also needs to secure his master key as its loss will not allow the user to recover data.

To mitigate these issues, the given paper proposes a new construction called "Dekey". It supports both file-level and block-level deduplication. Initially file level deduplication is performed (baseline approach) and if file is found then user will perform Proof of Work with the storage cloud service provider for proving the ownership of the file and it skips the block level deduplication. If file duplication is not found then only block level deduplication will be performed.

The idea of proposed solution is to apply deduplication technique on convergent keys. Then by using Ramp Secret Sharing Scheme (RSSS), the secret shares are constructed for convergent keys and these shares are distributed among different independent key servers. When the user uploads the data for the first time, he has to compute and share these secret shares whereas all the subsequent users having identical data copies are not required to construct and store the shares again. In order to regenerate the convergent key, the user has to get secret shares by accessing small number of key servers with the help of authentication at the time of data recovery. Dekey achieves client-side deduplication along with the Proof of Ownership (POW) which allows users to confirm their ownership of data copies to the storage server.

Thus, Dekey approach incurs limited storage overhead of convergent keys and provides efficient and reliable key management mechanism while protecting the convergent keys and confidentiality of outsourced data. RSSS used in Dekey, incurs encoding/decoding overhead but it is negligible as compared to network transmission overhead in regular upload/download operations.

3.4. Proof of Storage with Deduplication

The two main concerns of cloud computing are security and efficiency. Proof of Data Possession (POD) and Proof of Retrievability (POR) are generally used to identify if the data of a client is meddled with or not and Proof of Ownership (POW) is applied to mitigate the storage of multiple copies of same data which in turn reduces the usage of storage space and network bandwidth and thus making the storage efficient. In this way, POD, POR and POW are separately used for security and efficiency purposes. In Zheng and Xu (2012), a notion of Proof of Storage with Deduplication (POSD) is proposed which aims to combine the functionality of POD, POR and POW for making cloud storage secure and effective at the same time. The data that is outsourced to the cloud contains lot of duplicate files and for this reason almost every cloud storage provider uses deduplication which helps to remove redundant data and stores only unique instance of data. But deduplication comes with certain limitations as it is prone to attacks such as a malicious user can claim data which might not belong to him and to prevent such attacks we use POW. For performance of POSD, certain requirements are predefined in the given paper.

The proposed solution involves three participants: Cloud Storage Server: which provides storage space to the users and stores deduplicated data, Cloud Storage Client: which uploads data to the cloud and Auditor: is a third party to whom the client may give permission to verify the integrity of the data uploaded to the cloud, also the client who has a duplicated file may act as an auditor. The functional definition of POSD concept includes four operations: Key generation, uploading of files, data integrity and auditing protocol and deduplication checking protocol. The concept of Provable Data Possession (PDP) discussed in Ateniese et al. (2007); it allows a client that has stored data at an untrusted server to verify that the server possesses the original data without retrieving it. In this paper, POSD is defined as combination of Provable PDP and POW because POR uses erasure coding for retrieving the data and it is costlier than PDP. POSD scheme is tested on Random Oracle model based on Computational Diffie-Hellman assumption and it is proven to be secured. Also, the POSD scheme is evaluated for its performance and security and its efficiency is also compared with PDP, POR and POW schemes.

The complexity for data pre-processing is significantly less for POSD when compared to PDP and POR but the overhead incurred in communication by POSD is higher than the one incurred in PDP and POR. However, the POSD scheme is a little less efficient than POW. But POW does not provide facility to audit the data of cloud storage whereas POSD provides security, efficiency and

auditing of the data. The future work of this paper includes removal of random oracle in the protocol without affecting the performance.

3.5. Keyword search on the encrypted data stored in cloud storage

As per [Internet Society G7 Secure Communication \(2019\)](#), appropriate legal authorization, legitimate law enforcement agencies can perform keyword search in encrypted data to detect and intercept encrypted communications between terrorists and other malefactors. For retrieving the data upload to cloud in effective manner, [Li et al. \(2015\)](#) proposes the “keyword search” on the encrypted data stored in cloud storage. It shows two different constructions for secure keyword search mechanism and also does the security analysis to demonstrate how secure and confidential this keyword searching scheme is. For efficient retrieval of data from cloud storage, keyword search scheme is used in this paper and the proposed system also assures the privacy and confidentiality of the encrypted data and the information provided in keyword.

The given system of cloud storage consists of three entities: data owner, users and cloud storage server. Also, the given system supports both file and block level deduplication and each file is associated with the tag which is constructed by the hash function of the data file. This tag is used to perform duplicate check on data files. While retrieving data from cloud server, the trapdoors make sure that the sensitive information underlying the keyword is not revealed to the cloud server and adversary. For achieving data integrity and data confidentiality, Message Authentication Code (MAC) and convergent encryptions are used respectively.

Two different constructions are done in this deduplication cloud storage system supporting keyword search: the basic construction and the construction based on bloom filter. The three main functions in these constructions are file uploading, keyword searching and file retrieving. The extensions of the proposed system includes applying the keyword searching scheme to block-level deduplication as in current paper it is only used in file-level deduplication. In addition, Proof of Ownership (POW) can also be applied to the given system for higher security. The security analysis of this system is done on the basis of the underlying building secure blocks such as MAC and convergent encryption and based on these; it is proven that the proposed keyword search scheme is secure.

3.6. Secure Deduplication in hybrid architecture cloud storage

Various approaches of secure deduplication are discussed and analysed in [Patil and Kulkarni \(2015\)](#). Different primitives used for security are: Symmetric Encryption– which uses a secret key to encrypt and decrypt the data, Convergent Encryption–which calculates the hash of given data block and uses that hash value as a key to encrypt and decrypt, Proof of Ownership– it allows user to give proof of their ownership of data files on cloud storage, Identification Protocol– it is used for the identification of the user and to avoid malicious users. There are five different security methods analysed in this paper and it is concluded that the schemes are secure and provides confidentiality of data by providing security against various outsider and insider attacks.

3.7. Data Deduplication across various versions

[Rahumed et al. \(2011\)](#) proposes ‘Fade Version’ which allows assured deletion of outsourced data along with the version control of backed up data. It also removes duplicate data across various versions and also protects the outsourced data by using cryptographic encryption. Since the present version-controlled backup system and assured deletion are not compatible with each other, the main aim of Fade Version is to make a single design in which

both assured deletion and version control can be compatible with each other. Fade Version is an extension of Cumulus which is a version controlled cloud backup system and a new cryptographic implementation is added along with it for the purpose of assured deletion. The user's data files are organized in various versions (snapshots) and for every version, its metadata is present.

The proposed design employs two-layered encryption to gain successful assured deletion. Here, the assured deletion of any one version will not have any effect on any other versions even though various versions have data dependency among them. Several system entities are involved in the construction of Fade Version. They are: Backup Storage, Backup Module, Object Database, Key Escrow System and Stat Cache. In the proposed design, every data file is associated with three different policies and by knowing the association of different files with different policies, Fade Version can restore a file or revoke a policy and its associated files.

This paper compares the Cumulus and Fade Version for evaluating the performance overhead in terms of time, monetary cost and space which is caused by implementing assured deletion along with version control. It is found that – both systems have almost equal backup and restore time, Fade Version depicts more space increment monthly and Fade Version's monetary overhead is negligible. The future work for this paper includes research on decreasing the number of keys for its easy management and storage.

3.8. Duplicate Less Encryption for Simple Storage (DupLESS and modified DupLESS)

[Keelveedhi et al. \(2013\)](#) for the purpose of protection of their data, clients generally tend to encrypt their data while uploading it to the cloud storage and for this purpose they use their own key. Using different keys for same data results in different cipher, a text of similar data files and makes data deduplication unsuccessful. Therefore, Message-Locked Encryption (MLE) is used to solve this issue but then it becomes prone to brute-force attacks and adversary can get access of the files which belongs to known set. Among various solutions proposed for secure deduplication, DupLESS (Duplicate Less Encryption for Simple Storage) is utilised which gives both secure deduplication solution along with resistance to brute-force attacks.

[Duan \(2014\)](#) proposed the work to overcome the limitations of the work done by [Keelveedhi et al. \(2013\)](#). They developed the new method to eliminate the need of Key Server and implemented the distributed method for the same. Based on experiments, a decentralized version of encrypted deduplication that attains the same security as DupLESS.

[Shin et al. \(2017\)](#) provided the server-aided encryption to a decentralized setting that consists of multiple Key Servers. It provides an inter Key Server Deduplication scheme using the concept of blind signature in which cloud service provider can perform Deduplication over encrypted test from different Key Servers within a tenant or across tenants. Apart from that, authors implemented their scheme with the parallelism supported by multi-core architecture computing devices using Python language and it achieves low latency which performs better than existing schemes.

In [Prajapati and Shah \(2014\)](#) proposed architecture, modifications are made to the already existing DupLESS system using blowfish. In this system, a key server is shared among the group of clients to get message-derived keys to realize deduplication along with protection against other external attacks. As compared to the AES which induce more computational time, the blowfish algorithm reduces this computational time to a great extent. The proposed system has been tested by encryption and decryption time taken and it is found that it is significantly smaller as compared to the system using AES algorithm. In the proposed system, dedu-

plication is done at storage server side. The future work of given paper includes deploying keyword search in the proposed system and combining service denial with limitation of rate.

3.9. Data Deduplication in Crime Investigation

Influence of technology has increased the amount of crime and investigation of constantly increasing crimes requires digital forensic analysis. This has led to increase in the volume of data to be processed and analysed. However, present systems of digital forensic demand more time and manpower. The process of gathering data and evidences from various sources and analysing them, results in more processing time for each case. One of the main factors responsible for this is the speed with which the data is acquired from source. The conventional data acquiring step includes connection of hardware device being investigated, to a forensic workstation through write-blocker. This reading speed consumes large amount of time and during this the workstation remains engaged in this single procedure even prior to the beginning of evidence analysis step. Along with this, day by day the volume of data stored on device from which evidence is collected is also growing.

Scanlon (2016) proposed a novel approach to solve this issue- it states the use of centralized database and deduplication technique to mitigate the problems caused by conventional method. Some of the added requirements for this proposed system includes – use of deduplication to remove deduplicated data to increase data acquisition time, centralizing the analysis of digital forensic evidence for cost saving purpose and it also ensures the availability of new techniques and technologies to each department of forensic workstation and lastly, to have the resources shared centrally among various countries for collaboration and speeding up the whole process of investigation and analysis.

The implementation of proposed system will take on cloud-based infrastructure or in a centralised data centre. Similar to conventional approach, the device containing potential evidence will be connected to the workstation and according to proposed solution; the hash value of each artefact to be collected will be calculated and matched against the values stored in local database. This solution mitigates redundancy in analysis of data and it alleviates the disk reconstruction process. There are several benefits of this solution and it efficiently combats the issues resulting from digital forensic backlog.

3.10. File level data Deduplication in Hadoop

Prajapati et al. (2017), Hadoop is widely used for applications like healthcare, education, crime and fraud detection, etc. It is found that when user “A” uploads the same file “F1” more than one time with same file name (let say “F1”), it doesn’t allow saving the same file. But when user uploads the same file content with different file name (let say “F2”) Hadoop allows uploading that file. In general same files are uploaded by many users (cross user) with different name with same contents so this leads to wastage of storage space. Hadoop does not provide proficient and enhanced data storage solution for duplicate files. So file level data deduplication solution with data integrity is provided in this paper.

3.11. Current Data Deduplication Schemes

Bai et al. (2020) proposed the deduplicated data integrity auditing scheme supporting the ownership modification. It provides the integrity of the data with dynamic access control. They also used the concept of secure identity-based encryption, ElGamal encryption and Randomized Convergent Encryption using C language.

He et al. (2020) proposed the Deduplication based on data popularity with encrypted data. They used the concept of attribute-

based encryption and bi-linear mapping using C++. Deduplication can be done in their work without third party help. Premkamal et al. (2020) provided enhanced attribute based access control with secure Deduplication in which the cost of communication and computation overhead of deduplication and encryption is less compare to other schemes using Python language.

Lee and Park (2020) proposed novel blockchain based Deduplication to provide privacy of the CCTV video storage. Original CCTV video is encrypted and privacy masking levels are applied. The part of the CCTV video modification is identified by the synchronization procedure discussed in the paper. In this work sniffing attack is not possible because they are using different keys for data blocks. The main advantage of proposed SM-Tree over Merkle-Tree is; it allows data insertion or deletion in particular block. Their work supports variable size of data block which provide better effect in Deduplication.

Li et al. (2020) proposed Tunable Encrypted Deduplication, which identifies the duplicate chunks/blocks while Sketch based frequency counting. They also used probabilistic key generation in which provide the protection of information leakage of the same files by using candidate set of keys for encryption of chunks/blocks. Based on their experiments by effectively using C, C++ and Python languages, they achieved 50% Deduplication in File Storage Lab at Stony Brook University and achieved 65.72% of the Deduplication from Microsoft Windows File snapshots. Their work do not support any optimizations for restoring of files.

Li et al. (2020) provided the solution of information leakage in encrypted Deduplication using frequency analysis for multiple users using shared Duplicate data and to solve their previous work’s Li et al. (2020) limitations. Based on their experiments, they achieved 83.2% Deduplication in File Storage Lab at Stony Brook University and achieved 97.4% Deduplication from 156 virtual machine snapshots.

Liu et al. (2020) provided a novel Data Deduplication with verifiable attribute-based keyword search in encrypted data with the help of third party auditor. They used the bi-linear mapping and attribute-based encryption. Based on the experiments using C language, the communication overhead is low compare to other schemes. They kept dynamic file updating issue for their future work.

Nayak and Tripathy (2020) theoretically and practically achieved the better performance for secure server-aided Data Deduplication. Their work supports intra Key Server and cross Key Server Deduplication. Data confidentiality, Possession Proof, resistant against tag inconsistency attack and scalability were their goal of providing the Data Deduplication solution. To achieved the desired solution they used the symmetric encryption, bi-linear mapping and proxy re-encryption.

Shen et al. (2020) proposed a method to overcome the brute-force dictionary attacks which deals with the most crucial issue that is data leakage. They developed a lightweight method to provide Deduplication with strong privacy protection. The concept of Provable Data Possession (PDP) discussed in Ateniese et al. (2007) is used in their work. PDP allows a client that has stored data at an untrusted server to verify that the server possesses the original data without retrieving it. Using C language, they have done the experiments and compare their method with existing schemes. Based on their experiments and result discussion it is observed that their storage overhead cost, authenticator generation cost, proof generation cost, challenge generation cost and proof verification or auditing cost is more efficient.

Yin et al. (2020) provided MUti-tiered and SLA drivEn deduplication (MUSE) for cloud storage systems. They developed the novel Dedup-SLA (Deduplication oriented Service Level Agreement) which generally deals with the legal agreement between Cloud Service Providers and the cloud service consumers based on the

different Quality of Services (QoS) parameters defined in the legal agreement document. Then based on the advantages of different Deduplication schemes the system select appropriate combinations. After that authors implemented a method called Dynamic Deduplication Regulation (DDR) which adjust the deduplication behavior during runtime. They implemented and compared their scheme with twenty different providers (IBM, Microsoft, NetApp etc.) by experiments and based on the observation they found that their system works efficiently with the following parameters of the Deduplication like inline, offline, chunk-level, file-level, local, global, primary, configurable and dynamic.

Yuan et al. (2020) worked on a most important issue of Cloud Service Provider changes the integrity of consumers data. It is possible to have lots of duplicate copies of data among multiple users in the cloud storage. Authors used Ethereum blockchain for their experiments. They used the combination of blockchain and public key based homomorphic linear authenticator algorithm for verifying the data integrity without any help of third party and automatic punishment. They also used Hash and Convergent Encryption scheme developed by Bellare et al. (2013). Based on the Java and Solidity languages experiments, it is observed that their scheme is efficient in terms of probabilistic audit, batch auditing, privacy-preserving, Data Deduplication and fair arbitration.

Standard Attribute-based encryption system does not support the secure Data Deduplication. Wang et al. (2020) proposed a method to provide the same. Authors used the concept of hybrid cloud in which private cloud finds the Duplicate data and public cloud manage the storage. So instead of the sharing of the keys multiple users will access shared data based on the access policies. They used the concept of bi-linear pairing with symmetric encryption and Elliptic Curves using Python language for experiments and achieved the desired task of providing secure Deduplication with Attribute-based encryption.

4. Potential Directions and Ideas for Improvements

With today's focus on centralize data storage of cloud platform; data possession using Identity Based Encryption Kadvani et al. (2019), Additive Homomorphic Encryption Shah and Prajapati (2020) and file level and block level data deduplication are driving force in the research of information security and distributed data storage technologies. Significance of integrity and deduplication in the area of cloud computing data storage for the static as well as dynamic data possession with data deduplication would be explored.

An important part of this work is to create adaptive and portable integrity framework for cloud computing storage applications so that it can be used for integrity checking and provide solution of data deduplication. The following potential direction as objectives/scopes could be considered while designing the solution:

- To develop integrity framework and study the effect of performance with existing cloud computing data storage application.
- To verify integrity of data without downloading it from server by client.
- To reduce the computation complexity, that refers to the initialization and verification overheads at the client and the proof generating overheads on the server. It means that the scheme should be efficient in terms of computation.
- To reduce the communication overhead between client and server required by the scheme.
- To provide end-to-end (client and server both) data integrity verification.
- To provide adaptive and portable security framework for various cloud computing applications.

- To reduce the storage cost that refers to the additional storage of client and server required by the scheme. It means that the additional storage should be as low as possible.
- To provide quick response of data integrity query (though symmetric approach).
- To provide compatibility with data dynamics (insertion, deletion, updation operations).
- To maintain data privacy while assessing integrity (through symmetric encryption).
- To allow private verifiability, public verifiability and batch auditing.
- To reduce the execution time while providing confidentiality of the data.
- To provide efficient key generation, management and key distribution.
- To provide secure and efficient data deduplication with both file level and block level with least cost.
- To provide secure data deduplication with searchable encryption for detecting and intercepting encrypted communications between terrorists and other malefactors.
- Apart from that, we can use the concept of file data deduplication and searchable encryption in real life mobile applications like WhatsApp (2020); Telegram (2020); Viber (2020) and other applications.

5. Conclusion

In this literature review, various approaches for providing secure deduplication in cloud storage are discussed. The different schemes proposed above can be employed with existing cloud storage services. The security analysis of various schemes mentioned above guarantees the security and integrity of client's data on cloud from both insider as well as outsider attacks. By combining various security approaches with data deduplication, the security concerns of clients are addressed along with elimination of redundant data from cloud storage which improves storage efficiency and reduces upload bandwidth. However, there are certain issues which remain to be resolved. One of them is storage and management overhead caused by cryptographic keys also the issue of breach of client's key is yet to be solved. Moreover, the above-mentioned schemes discuss about storage and retrieval of data from cloud storage but in case of single user and multiple users (cross user) it's more difficult to the implement edit and deletion operations with secure data deduplication solution. Based on above study it is found that block level and byte level data deduplication consume more time for implementation so generally file level data deduplication is used. The combination of file level and block level Deduplication can be used to provide better efficiency. Initially file level deduplication can be performed and if file is found then user can perform Proof of Work with the storage cloud service provider for proving the ownership of the file and it can skips the block level deduplication. If file duplication is not found then only block level deduplication can be performed. One can provide data deduplication solution at client side and server side based on the applications. If data deduplication is provided within small organization within private cloud storage then client side data deduplication can be used to minimize the storage server computation, bandwidth and time can be saved. The concept of Attribute Based Encryption, Identity Based Encryption can be combined to provide secure Data Deduplication to provide better security with less overhead.

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